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Physiological responses to feed restriction and starvation in juvenile Siberian sturgeon *Acipenser baerii* (Brandt, 1869): Effects on growth, body composition and blood plasma metabolites

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Abstract

An 8-week study was applied to evaluate the effects of feed restriction and starvation on growth, proximate composition, haematological and biochemical factors in juvenile Siberian sturgeon *Acipenser baerii* with an average weight of 54 ± 0.6 g. Fish were randomly divided into nine indoor tanks (1,050 L volume) with 30 fish per tank in a flow-through system at $18 \pm 0.4^\circ\text{C}$. Three feeding strategies were considered: I) fed at satiation (F), II) restricted feeding at the rate of 1% body weight per day (R) and III) starvation (S) each with 3 replicates. Fish were fed with commercial pellets 4 times daily, but no feed was considered for S group throughout the trial. Basal level at the start, 4 and 8 weeks after initiation of the experiment, blood samples were taken from 5 fish/tank to measure the haematological and biochemical parameters. Moreover, growth performance was recorded every 2-week interval. At the end of the experiment, the growth parameters and proximate composition were significantly different among the treatments. As expected, the highest weight (164.7 ± 2.1 g) was observed in F group compared to R (101 ± 1.7 g), or S (42.4 ± 0.8 g) groups. The highest ($12.07 \pm 0.14\%$) and lowest ($5.37 \pm 0.42\%$) fat contents were observed in F and S groups, respectively, while the protein was not affected in R group compared with F group. In addition, most haematological and biochemical indices of Siberian sturgeon were affected by starvation at week 4 and 8, while few changes were observed in restricted feeding. The highest mean corpuscular volume, mean corpuscular haemoglobin, plasma total protein, cholesterol and triglyceride was found in S group, while the lowest value of plasma glucose was observed in this group. It can be concluded that Siberian sturgeon adjusts the physiological condition in short-term (4 weeks) feed restriction using stored metabolites to have basal metabolism and due to this, a short period of feed restriction strategy recommends to the optimum nutritional management of Siberian sturgeon to cost-saving during unsuitable conditions.

KEYWORDS

biochemical indices, body composition, fasting, growth, haematology, restricted feeding, sturgeon

1 | INTRODUCTION

In many ecosystems living organisms encounter with food deprivation period during their life cycle. Some fish species can tolerate long periods of feed deprivation in natural conditions (Navarro & Gutiérrez, 1995; Stepanowska, Nędzarek, & Rakusa-Suszczewski, 2006) as well as in aquaculture operations (Furné et al., 2008, 2012; Xiao et al., 2011). In the natural environment, stress conditions such as pollution, wintering phases, long period of environmental fluctuations, changes in food availability, water temperature and preparation for spawning are the main reasons for starvation occurrences. In aquaculture condition, stressors such as pathogens, low water quality and temperature fluctuations also affect feed intake and growth performance (Falahatkar, 2012). Because of this, for better management farmers use various feeding strategies such as starvation or feed restriction to decrease production costs (Falahatkar, Akhavan, Efatpanah, & Meknatkhah, 2013; Urbinati, Sarmiento, & Takahashi, 2014), improve water quality (Davis & Gaylord, 2011), reduce mortality (Shoemaker, Klesius, Lim, & Yildirim, 2003), saving personnel time and labour cost (Ali, Nicieza, & Wootton, 2003; Falahatkar, 2012; Krogdahl & Bakke-McKellep, 2005).

In food deprivation periods, fish looks for optimal ways to access adaptive strategies for survival (Gillis & Ballantyne, 1996; McCue, 2010; Navarro & Gutiérrez, 1995). The strategies including converting endogenous energy reserves from tissues and mobilize body components in order to obtain the energy requirement (Furné et al., 2012; Pérez-Jiménez et al., 2012). This ability operates metabolic changes that are depended on species, fish age or nutritional status (Navarro & Gutiérrez, 1995; Wang, Hung, & Randall, 2006). Therefore, it is important to find how food deprivation periods might alters some biochemical changes in fish.

Sturgeons are valuable species which have been sharply declined in recent years and now are considered as endangered species (IUCN, 2016). Siberian sturgeon *Acipenser baerii* is the most commonly farmed Acipenserid species in Europe and Asia (Falahatkar, 2018). Among the sturgeons, this species has been considered as a biological model for nutritional and physiological studies, because of easy food adaptation and reach early puberty age (Eslamloo, Falahatkar, & Yokoyama, 2012; Falahatkar, 2018). These species like many other temperate fish species occasionally may face fasting as a result of natural alternations (Gillis & Ballantyne, 1996) or during the culture period or releasing time for stock rehabilitation (Falahatkar, 2012).

Although Siberian sturgeon has attracted more attention around the world, haematological parameters as physiological indicators have not been extremely examined under different rearing or feeding conditions. The lack of information on the effect of feed deprivation or restriction on the growth performance and haematological parameters of juvenile Siberian sturgeon led to investigate this issue. Therefore, the aim of the present study was to evaluate the haematological and physiological changes including the concentrations of plasma glucose, triglyceride, cholesterol, total protein and proximate composition of Siberian sturgeon in response to a period of fasting or restricted feeding strategies.

2 | MATERIALS AND METHODS

2.1 | Experimental design and feeding strategies

Two hundred and seventy-five Siberian sturgeon juveniles with 54 ± 0.6 g (mean $\pm$ SE) wet weight and 25.3 ± 0.1 cm total length were obtained from the Shahid Dr. Beheshti Sturgeon Fish Propagation and Rearing Complex (Rasht, Gilan, Iran). The fish were adapted to experimental condition 2 weeks prior to starting the trial. During this period, all fish were fed ad libitum four times daily at 8:00, 12:00, 16:00 and 20:00 (Biomar, 48% crude protein and 22% crude fat). Fish were randomly divided into three treatments included nine indoor fibreglass tanks (1,050 L volume) with 30 fish per tank in a flow-through system with water supplied from the river mixed with well water (approximately with the ratio of 1:1) to prevent a sharp drop of temperature during the fall season. Water flow rate in each tank was maintained at 14.4 ± 0.2 L/min. Fish were kept at a low-density condition to avoid stress during the experiment with the initial density of 1.5 g/L. Water temperature was $18 \pm 0.4^\circ\text{C}$, dissolved oxygen 7.1 ± 0.2 mg/L and pH 7.7 ± 0.1 throughout the trial under 12L:12D photoperiod.

Three feeding strategies were applied as; (a) fed group, where fish was fed at satiation for 8 weeks (F), (b) restricted feeding at the rate of 1% body weight daily for 8 weeks (R), and (c) starved for 8 weeks (S). Feed supply was adjusted as time progressed every 2 weeks after measuring of weight as fish increased in size. Fish in the fed treatments (F and R) were hand fed with commercial pellets (Biomar, Nersac, France, No. 1.9; 48% crude protein, 22% crude fat, 8% ash and 2% fibre) 4 times daily at 8:00, 12:00, 16:00 and 20:00. At feeding groups, fish were fed to satiation, until that the fish stopped feeding actively. Restricted ration was given also as 4 meals per day, each of 0.25% tank biomass. The duration of each meal was around 30 min. Uneaten feed was drained and feed intake was evaluated for each tank.

2.2 | Growth performance

To assess the growth performance, body weight (to the nearest 0.01 g) and total length (to the nearest 1 mm) of all fish were measured every 2 weeks interval. Biometric data were recorded following that all fish were starved for 24 hr to evacuate their gut before measuring. The fish were anaesthetized with 0.5 g/L clove powder and after weighing and assess the length, fish were returned to their tank for recovery. At the termination of the experiment, weight gain (WG), condition factor (CF), specific growth rate (SGR), feed intake (FI), feed conversion ratio (FCR), viscerosomatic index (VSI) and hepatosomatic index (HSI) were calculated as follows:

$$\text{WG (\%)} = 100 \times (\text{final body weight} - \text{initial body weight}) / \text{initial body weight}$$

$$\text{CF} = 100 \times (\text{body weight} / \text{total length}^3)$$

$$\text{SGR (\% day}^{-1}\text{)} = 100 \times (\text{Ln final body weight} - \text{Ln initial body weight}) / \text{days of the trial}$$

FCR = dry feed intake/weight gain

FI (g fish⁻¹) = feed intake/number of fish

HSI (%) = 100 × (liver weight/body weight)

VSI (%) = 100 × (viscera weight/body weight)

To calculate the HSI and VSI, 5 fish from each tank were gently killed by an overdose of clove powder and viscera or liver weights were recorded at the nearest 0.01 g. All fish husbandry, manipulations and samplings were conducted in accordance with local animal ethics guidelines.

2.3 | Proximate composition

Before the initiation of the experiment, 5 fish were considered for initial proximate body composition. At the end of the trial, fifteen fish per group (five fish in each replicate) were killed and immediately frozen at -20°C for whole body proximate composition. Fish were ground prior to proximate analysis and whole body was homogenized for each fish. Samples were placed in oven drying at 105°C to reach constant weight to find the moisture content. Crude fat was measured by Soxhlet method, based on the extraction of the lipids (AOAC, 1996). Crude protein content was evaluated by Kjeldahl method by measuring the total nitrogen (N × 6.25). To measure the ash content, samples were burned for 6 hr in a Gallenkamp furnace at 550°C.

2.4 | Blood collection and haemato-biochemical analysis

The blood samples of five fish were randomly collected from each replicate at the beginning, middle (week 4) and termination of the experiment (week 8). The blood was taken from the caudal vessel using a 2 ml heparinized syringe to specify haematological and biochemical parameters. The sampling lasted less than 1 min for each fish after anaesthetization. Samples from each fish were treated separately. An aliquot of whole blood was transferred to an Eppendorf tube and used for determination of haematocrit (Hct) value, haemoglobin (Hb) concentration, numbers of red blood cells (RBC) and white blood cells (WBC). Afterwards, the mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH) and mean corpuscular haemoglobin concentration (MCHC) were measured according to below formulas (Klinger, Blazer, & Echevarria, 1996):

MCH (pg cell⁻¹) = 10 × (Hb/RBC)

MCV (fL) = 10 × (Hct/RBC)

MCHC (%) = 100 × (Hb/Hct)

The blood was centrifuged in heparinized microhaematocrit capillary tubes after centrifugation in a standard microhaematocrit centrifuge at 3,500 g for 7 min to determine Hct values. Hb concentration was measured using a commercial kit (Cyanmethaemoglobin method; Sigma). Numbers of RBC and WBC were counted with an improved Neubauer haemocytometer (Houston, 1990).

The second blood aliquot was centrifuged at 1,600 g for 10 min, then plasma separated and stored in Eppendorf tubes at -80°C for subsequent assays of biochemical factors including glucose, total protein, triglyceride, and cholesterol concentrations.

Plasma glucose levels were determined by glucose-oxidase (GOD-PAP) method (Bayunova, Barannikova, & Semenkova, 2002) using a commercial kit (Pars Azmun diagnostics kit). Total protein, triglyceride and cholesterol concentrations were determined by Biuret method (Pars Azmun diagnostics kit), GPO-PAP (MAN diagnostics kit) and CHOD-PAP (MAN diagnostics kit) methods, respectively, in accordance with Fynn-Aikins, Hung, Liu, and Li (1992).

2.5 | Data analysis

Homogeneity of variances and normality of data were checked by Levene's and Kolmogorov-Smirnov tests, respectively. To compare changes in growth performance in each step of biometry between the groups and also proximate composition at the end of the trial, data were analysed by one-way analysis of variance (ANOVA). Haemato-biochemical data were subjected to two-way ANOVA to determine differences between treatments and times in all sampling steps. Duncan's multiple range test was performed to find any difference among the means. The level of significance was chosen at $p < .05$. All analyses were performed using SPSS software (SPSS, 16.0). Data are presented as mean ± standard error (SE).

3 | RESULTS

3.1 | Growth performance

The growth parameters are shown in Table 1. The body weight was significantly different ($p < .05$) throughout the experiment among the treatments (Figure 1). After 8-week trial, the WG of F and R groups increased 201.6% and 89.6%, respectively, and decreased 21.6% in starved fish. Body length and CF also were significantly different in all groups ($p < .05$). SGR showed significant differences ($p < .05$) during the experiment; F and R groups had positive SGR following the feeding and negative in S group. FI was significantly higher in the F group than the R group. FCR in the R group was improved compared with the F group. The HSI of the F group was significantly higher than the other feeding regimes ($p < .05$). VSI of the F and R groups was significantly higher than the S group ($p < .05$), but no difference was found between the F and R groups.

3.2 | Proximate composition

The proximate carcass composition at the end of the experiment is shown in Table 2. Significant differences ($p < .05$) were observed among the groups for moisture and crude fat, but no changes were found in ash content ($p > .05$). Moreover, crude protein content in S group decreased significantly compared with other groups ($p < .05$).

TABLE 1 Changes in final weight, final length, weight gain (WG), condition factor (CF), specific growth rate (SGR), feed intake (FI), feed conversion ratio (FCR), hepatosomatic index (HSI), viscera somatic index (VSI) and survival rate in juvenile Siberian sturgeon *Acipenser baerii* after 8 weeks of applying different feeding strategies (F: full-fed, R: restricted feeding and S: starvation)^a

	Feeding strategies		
	F	R	S
Initial weight (g)	54.6 ± 0.6	53.3 ± 0.7	54.1 ± 0.7
Final weight (g)	164.7 ± 2.1 ^a	101.0 ± 1.7 ^b	42.4 ± 0.8 ^c
Initial length (cm)	25.4 ± 0.1	25.1 ± 0.1	25.4 ± 0.1
Final length (cm)	36.6 ± 0.2 ^a	32.4 ± 0.2 ^b	26.1 ± 0.2 ^c
WG ^b (%)	201.6 ± 4.2 ^a	89.6 ± 3.2 ^b	-21.6 ± 3.0 ^c
CF ^c	0.34 ± 0.00 ^a	0.30 ± 0.01 ^b	0.27 ± 0.01 ^c
SGR ^d (% day ⁻¹)	1.9 ± 0.01 ^a	1.1 ± 0.01 ^b	-0.42 ± 0.03 ^c
FI ^e (g fish ⁻¹)	187.1 ± 31.2 ^a	42.9 ± 4.8 ^b	-
FCR ^f	1.7 ± 0.3 ^a	0.9 ± 0.1 ^b	-
HSI ^g (%)	4.2 ± 0.2 ^a	2.7 ± 0.2 ^b	0.8 ± 0.1 ^c
VSI ^h (%)	11 ± 0.3 ^a	9.7 ± 0.5 ^a	6.6 ± 0.4 ^b
Survival rate (%)	100	100	98.8 ± 1.1

^aValues are mean ± SE from all fish of each treatment, and 15 fish/treatment for HSI and VSI (n = 3). Different superscripts in the same row indicate significant differences ($p < .05$) between feeding strategies by one-way ANOVA and Duncan's test.

^bWG = $100 \times (\text{final body weight} - \text{initial body weight}) / \text{initial body weight}$.

^cCF = $100 \times (\text{body weight} / \text{total length}^3)$.

^dSGR = $100 \times [\ln(\text{final body weight}) - \ln(\text{initial body weight})] / \text{days of trial}$.

^eFI = feed intake/number of fish.

^fFCR = dry feed intake/weight gain.

^gHSI = $100 \times (\text{liver weight} / \text{body weight})$.

^hVSI = $100 \times (\text{viscera weight} / \text{body weight})$.

3.3 | Haematological parameters

Haematological parameters were significantly affected by feeding strategies and duration of the trial ($p < .05$). WBC was influenced by long-term feed deprivation and restricted feeding compared to full-fed fish ($p < .05$). WBC value showed reduction throughout the experiment, with the highest and lowest values at the initiation of the experiment ($11,720 \pm 1,089 \text{ cell/mm}^3$) and in starved

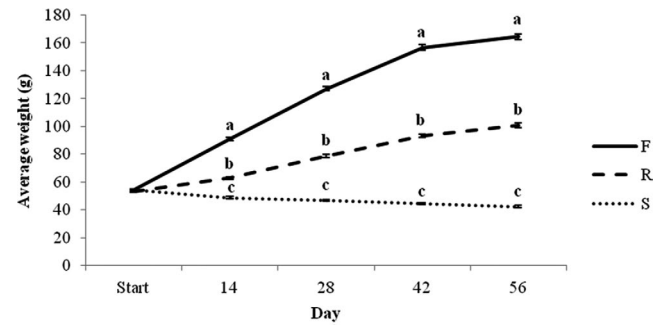


FIGURE 1 Changes in average weight (g) during 8 weeks of applying different feeding regimes (F: full-fed, R: restricted feeding and S: starvation) in juvenile Siberian sturgeon *Acipenser baerii*. Values are mean ± SE from all fish of each treatment (n = 3). Different superscripts indicate significant differences ($p < .05$) between feeding strategies by one-way ANOVA and Duncan's test

fish at the end of the experiment ($4,358.3 \pm 209 \text{ cell/mm}^3$) respectively ($p < .05$; Figure 2A). No significant differences in RBCs and Hb values were found between groups, although in each group decreased during the trial ($p < .05$). In all groups, lowest values were observed at the end of the experiment ($p < .05$), while until the week 4, despite changes in RBCs and Hb, differences were not significant (Figure 2B,C). Hct values were affected by long-time feeding strategies ($p < .05$; Figure 2D). The lowest value ($18.5 \pm 0.9\%$) was found in starved fish at the end of trial with a 38.9% decrease compared to the initial value ($p < .05$). Although MCV and MCHC values were affected by the long-time feeding strategies ($p < .05$; Figure 2F,G), MCH showed difference during the time in R group, but not changed between treatments at the middle of experiment ($p > .05$; Figure 2E).

3.4 | Biochemical parameters

A significant changes in plasma glucose concentrations were recorded only after 8 weeks, but in all treatments decreased significantly ($p < .05$; Figure 3A). In starved fish, glucose concentration decreased after second sampling (22.6% of the initial value) and was significantly lower than the R and F groups at the end of the trial ($p < .05$), but restricted-fed fish did not show considerable differences with full-fed fish.

TABLE 2 Changes in proximate carcass composition (as % of wet weight) in juvenile Siberian sturgeon *Acipenser baerii* after 8 weeks of applying different feeding strategies (F: full-fed, R: restricted feeding and S: starvation)^a

	Feeding strategies			
	Initial	F	R	S
Moisture (%)	76.40 ± 0.2	71.75 ± 0.12 ^c	75.15 ± 0.23 ^b	80.94 ± 0.43 ^a
Fat (%)	9.24 ± 0.06	12.07 ± 0.14 ^a	8.89 ± 0.23 ^b	5.37 ± 0.42 ^c
Protein (%)	11.81 ± 0.97	12.45 ± 0.06 ^a	12.43 ± 0.05 ^a	10.13 ± 0.39 ^b
Ash (%)	2.43 ± 0.07	2.76 ± 0.04	2.64 ± 0.03	2.62 ± 0.13

^aValues are mean ± SE from all fish of each treatment and 15 fish/treatment for proximate composition (n = 3). Different superscripts in the same row indicate significant differences ($p < .05$) between feeding strategies by one-way ANOVA and Duncan's test.

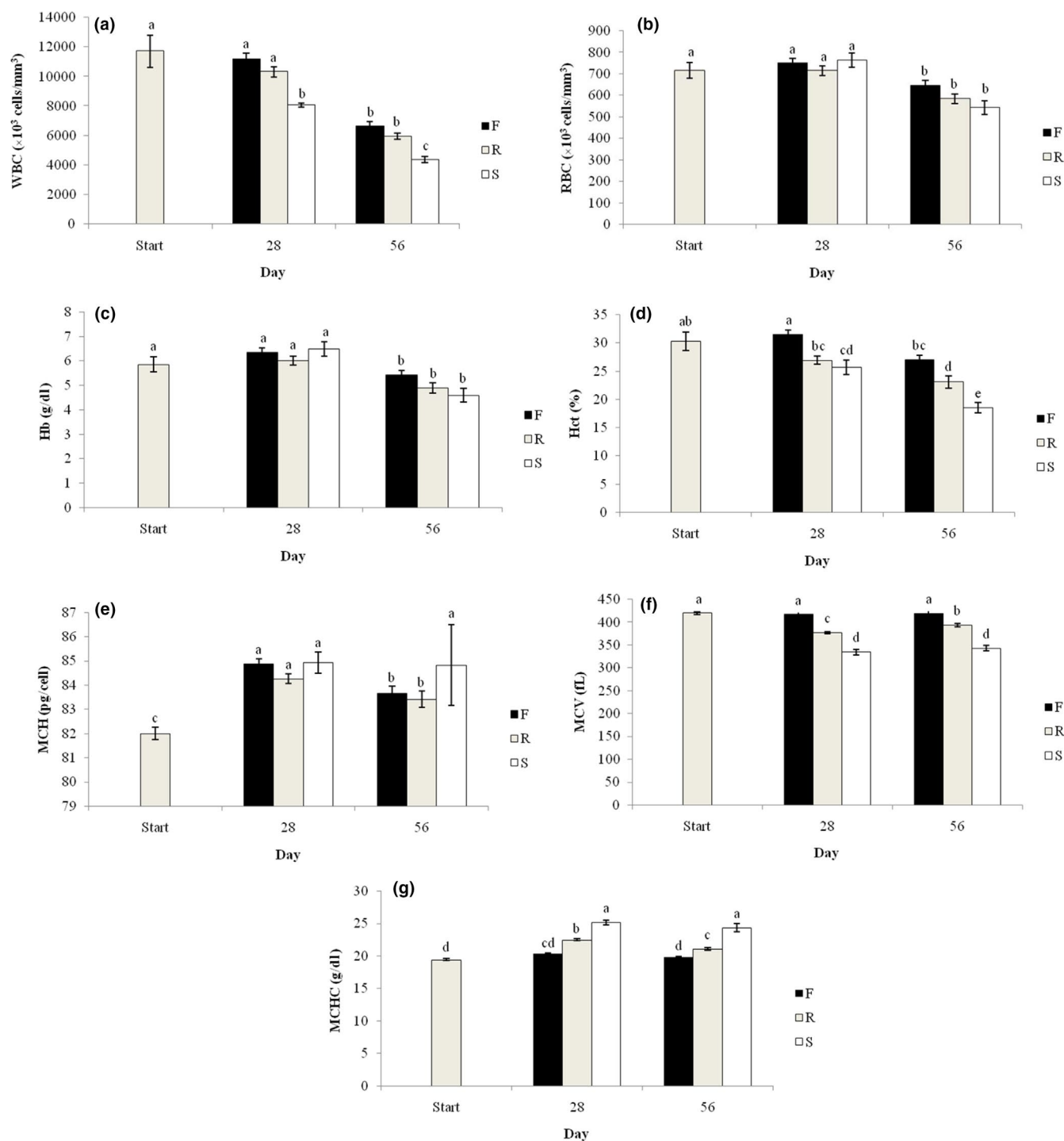


FIGURE 2 Changes in (a) number of white blood cells (WBCs), (b) number of red blood cells (RBCs), (c) haemoglobin (Hb), (d) haematocrit (Hct), (e) mean corpuscular haemoglobin (MCH), (f) mean corpuscular volume (MCV) and (g) mean corpuscular haemoglobin concentration (MCHC) during 8 weeks of applying different feeding regimes [F: full-fed (black column), R: restricted feeding (grey column) and S: starved (white column)] in juvenile Siberian sturgeon *Acipenser baerii*. Values are mean $\pm$ SE from 15 fish/ treatment ($n = 3$). In fig a–d, A and B letters indicate significant differences ($p < .05$) between feeding strategies in the same time and a, b and c letters indicate significant differences ($p < .05$) in each feeding strategies during the time, and in figs e–g, different letters indicate significant differences ($p < .05$). Differences are indicated by two-way ANOVA and Duncan's test [Colour figure can be viewed at wileyonlinelibrary.com]

Plasma total protein levels showed changes in each treatment during the time (Figure 3B), which was significantly decreased in the F group to the end of the trial, and lowest level (3.5 ± 0.1 g/L) was found at the end of the trial. After 4 and 8 weeks, total protein

was significantly higher in the S group compared with other groups (11.8 ± 0.9 g/L after 4 weeks and 5.7 ± 1.9 g/L after 8 weeks; $p < .05$). In the R and S groups, also significant decrease was found from onset to the end of the trial ($p < .05$).

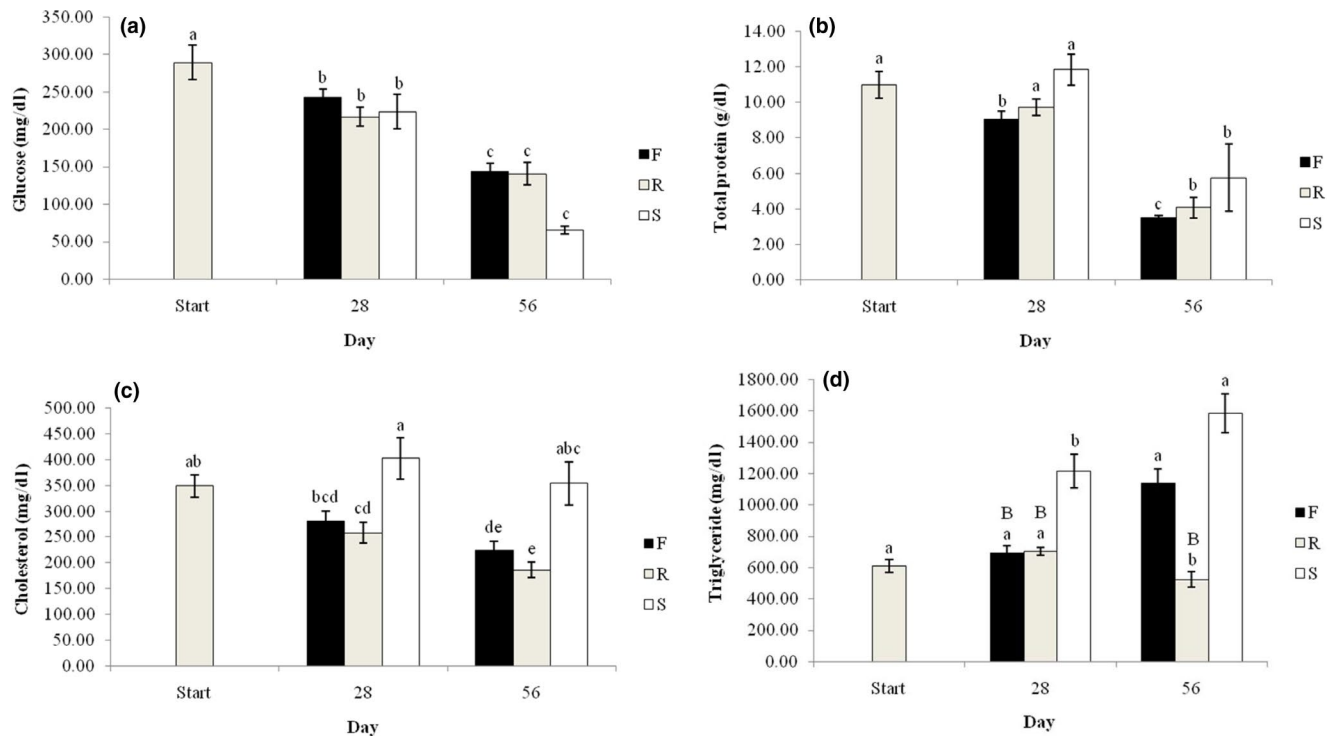


FIGURE 3 Changes in (a) glucose, (b) total protein, (c) cholesterol and (d) and triglyceride during 8 weeks of applying different feeding regimes [F: full-fed (black column), R: restricted feeding (grey column) and S: starved (white column)] in juvenile Siberian sturgeon *Acipenser baerii*. Values are mean $\pm$ SE from all fish of each treatment and 15 fish/ treatment ($n = 3$). In figure a and b, different letters indicate significant differences ($p < .05$) in each feeding strategies during the time, and in figure c and d, different letters indicate significant differences ($p < .05$). Differences are indicated by two-way ANOVA and Duncan's test [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Plasma cholesterol and triglyceride concentrations were affected by trial duration and feeding strategies ($p < .05$). The highest levels of cholesterol were detected in the S group in mid and end of the trial ($p < .05$; Figure 3C). Triglyceride increased significantly throughout the trial, and highest levels were found at the end of 8 weeks ($p < .05$; Figure 3D).

4 | DISCUSSION

The present study indicated that which strategy would be better for long-term food deprivation periods in Siberian sturgeon. Unfortunately, poor and inconsistent knowledge is available about the growth and physiological responses of sturgeon during food deprivation period. As expected, feed deprivation and restricted feeding for 8 weeks showed significant changes in growth performance and proximate body composition in juvenile Siberian sturgeon that caused a negative impact on body weight, SGR, BWI, CF, HSI and VSI. According to proximate composition, fish used lipid (41.8% and 3.7% fall in the S and R groups, respectively) and fewer protein reserves (14.2% fall compared to initial content in starved fish) to provide required energy during the food deprivation periods.

Similarly, weight loss has been reported in white sturgeon *A. transmontanus*, beluga *Huso huso*, Persian sturgeon *A. persicus* and Siberian sturgeon after short or long period of feed deprivation (Falahatkar,

2012; Hung, Liu, Li, Storebakken, & Cui, 1997; Jafari, Falahatkar, & Sajjadi, 2018; Yarmohammadi et al., 2012) as well as reports in teleost fish (Caruso et al., 2012, 2010; Peres, Santos, & Oliva-Teles, 2011). Nevertheless, lack of considerable weight loss was reported in lake sturgeon *A. fulvescens* (Gillis & Ballantyne, 1996) and also in some other teleosts (Krogdahl & Bakke-McKellep, 2005). In fact, weight loss over the time resulted from reducing the metabolism rate during starvation (Hung et al., 1997) and due to utilizing of internal deposits in this situation as well as loss of muscle mass because of intense proteolysis which could cause total weight loss. However, previous studies stated that lack of weight loss can be due to replacement of energy sources of tissues by water to maintain the mass of starved fish (Navarro & Gutiérrez, 1995). Considering all these arguments, growth responses are in relation to species, age and duration of this period as well as the individual traits (Caruso et al., 2010).

In Chinese sturgeon *A. sinensis*, decrease in HSI occurred after 2 (Xiao et al., 2011) and 7 days of starvation (Liu et al., 2011). In Siberian sturgeon, decrease in HSI and VSI was found with an increasing length of starvation period (Ashouri et al., 2013). Decrease in HSI also has been reported in some teleosts (Ali et al., 2003; Caruso et al., 2012; Peterson & Small, 2004). According to the present study and previous reports, HSI and VSI reduction in food deprivation reflects the key role of liver and visceral deposits as energy reserves during starvation (Ashouri et al., 2013; Davis & Gaylord, 2011; Power, Melo, & Santos, 2000) and lack of nutrients transfer

into the liver by biochemical adjustments (Davis & Gaylord, 2011). The lack of significant difference in VSI in restricted-fed fish and control along with differences in HSI demonstrates liver store preference versus the muscle reserves to mobilization in food deprivation period.

As mentioned, proximate body composition analysis showed that in starved and food restriction fish changes in protein content were lower than the lipid, but in restricted feeding, no considerable changes occurred in carcass protein content. It seems that the lipid stores as first energy fuel consumed during the food shortage in Siberian sturgeon, and then, protein sources mobilized. Similar results have been reported previously in other sturgeon species (Hung et al., 1997; Jafari et al., 2018; Liu et al., 2011; Xiao et al., 2011) and some teleosts (Abdel-Tawwab, Khattab, Ahmad, & Shalaby, 2006; Davis & Gaylord, 2011; Tian, Fang, & Dong, 2010) confirmed our results. Along with the reduction in lipid and protein reserves, tissue water content is increased in most reports confirm our findings (Falahatkar, 2012; Hung et al., 1997; Liu et al., 2011; Xiao et al., 2011), because of an increase in tissue hydration rate (Ali et al., 2003) and osmotic pressure due to increase in metabolite levels during this period (McCue, 2010).

In restricted-fed fish, FCR has improved considerably compared to control. Cacho, Hatch, and Kinnucan (1990) reported although the higher growth occurred in the satiated group, best FCR was shown in lower feed rations. It seems that limited-fed fish have enough time for food digestion and absorption while they had lower FI, and for this reason, they showed the higher efficiency in relation to producing flesh than satiated feeding. In other words, high food availability in the control leads to inefficient digestion and absorption of nutrients in feed intake. So, despite the satiation feeding in control, FCR has increased and the food efficiency has reduced than the limited-fed group.

Starvation and restricted feeding strategies can be considered as a chronic stress situation and have considerable effects on fish physiology and biochemistry. In our study, the number of WBCs was similar to finding on rainbow trout (López-Luna, Vásquez, Torrent, & Villarroel, 2013) which showed the continuous decline in food deprivation period, that likely a reason that may be in relation to impairment of immune functions (López-Luna et al., 2013). The RBC count and Hb concentration were not affected by starvation, but Hct value decreased. Hct in sturgeon species has been displayed conflictive responses, including decrease in lake sturgeon (Gillis & Ballantyne, 1996) and increase in beluga (Falahatkar, 2012). Jafari, Falahatkar, and Sajjadi (2019) reported that except for the number of monocytes no changes were found in haematological parameters in Siberian sturgeon in 2 different sizes when exposed to different feeding strategies after 45 days. Similar to our findings, traíra (neotropical fish) *Hoplias malabaricus* (Rios, Kalinin, & Rantin, 2002; Rios, Oba, Fernandes, Kalinin, & Rantin, 2005) showed decrease in Hct value, whereas European eel *Anguilla anguilla* showed an increase in Hct during this period (Johansson-Sjöbeck, Dave, Larsson, Lewander, & Lidman, 1975). No changes in Hct have been reported in teleosts such as European sea bass *Dicentrarchus labrax*, blackspot seabream *Pagellus bogaraveo* (Caruso et al., 2011), sunshine bass *Morone*

chrysops × *Morone saxatilis* (Davis & Gaylord, 2011), red porgy *Pagrus pagrus* (Caruso et al., 2012) and rainbow trout (López-Luna et al., 2013). These anomalous responses may occur due to difference in species and experimental condition. It is possible that changes in haematology during the starvation period are influenced by erythropoietic reduction (Weinberg, Siegel, & Gordon, 1973) or changes in body water content (McCue, 2010). Decrease in Hb concentration in starved fish was related to diminishing blood metabolites and also declining blood osmotic pressure during the starvation period which eventually leads to reduction in MCV. This change could be due to decrease in the Hct. Hct reduction in concomitant with lack of significant changes in Hb redounded to increase in MCHC in the present study indicated that oxygen requirement of Siberian sturgeon was abided in restricted-fed and starved fish. In addition, lack of considerable changes in RBCs and MCH with reducing of MCV showed that the levels of Hb and oxygen carrying capacity of the blood remained in diminished RBCs.

Depletion of plasma nutrients is inevitable during starvation and restricted feeding. Our findings showed that glucose concentration decreased significantly only in starved fish after 8 weeks. Glucose is the main metabolite known to involve in important bioenergetic functions and leads to supply energy for tissues and cells by gluconeogenesis and glycogenolysis (Gillis & Ballantyne, 1996). Earlier studies have shown varied glucose responses in sturgeon species; consistent with our finding, white sturgeon, Adriatic sturgeon *A. naccarii* and Siberian sturgeon showed a significant decrease in plasma glucose level during starvation (Furné et al., 2012; Hung et al., 1997; Jafari et al., 2018). In contrast, the preserve of plasma glucose during starvation has been observed in lake sturgeon and Persian sturgeon (Gillis & Ballantyne, 1996; Yarmohammadi et al., 2012). In addition, many teleost species such as channel catfish *Ictalurus punctatus* (Peterson & Small, 2004), Nile tilapia *Oreochromis niloticus* (Abdel-Tawwab et al., 2006) and rainbow trout (Furné et al., 2012) are unable to maintain plasma glucose levels in extended starvation periods. However, constant glucose value was reported in several studies (Caruso et al., 2011, 2012; Davis & Gaylord, 2011; López-Luna et al., 2013). The mechanistic pathways of energy maintenance and other related metabolites should be clearly demonstrated in future research.

After 4 and 8 weeks of the experiment, plasma total protein was significantly higher in starved fish compared to other treatments, and in each group, this level was decreased during 8 weeks. Proteins found as fundamental plasma components that carry out the key functions in metabolism and physiology of organisms. Proteolysis happens when more comfily endogenous energy resources like glycogen and lipid are greatly utilized (Navarro & Gutiérrez, 1995). Depletion of plasma protein was found in starved beluga for 6 weeks (Falahatkar, 2012). No considerable changes were found in plasma protein levels in white sturgeon and Siberian sturgeon in long- and short-term starvation periods, respectively (Ashouri et al., 2013; Hung et al., 1997; Jafari et al., 2018). In Adriatic sturgeon, fluctuations of plasma total protein levels were observed during the starvation period (Furné et al., 2012). In teleost, plasma total protein levels

considerably decreased in Senegalese sole *Solea senegalensis* and Atlantic salmon *Salmo salar* (Costas et al., 2011; Hevrøy et al., 2011) during starvation. With lengthening of the starvation period due to discharge, the ready energy stores (glucose and lipids) and plasma total protein (as the ultimate energy reserve) partially used as spare reserves (Higginson, McNamara, & Houston, 2014) to maintain essential metabolism. Indeed, it seems that higher total protein level only in starved fish resulted from mobilization of these reserves and their release to blood circulation in the last stage of starvation in Siberian sturgeon, as previously mentioned carcass protein content also decreased just in starved fish. Fall of protein level in each group throughout the experiment may be reflecting temperature reduction during this trial. Such as in the F group, plasma glucose and total protein decreased significantly with the increase of feeding period. This could be due to the length of feeding/fasting period, size of fish and decreasing the water temperature in the last part of the experiment. Moreover, the decreased protein content could indicate elevated rate of proteolysis and/or increased amino acid oxidation associated with a non-compensatory rate of protein synthesis which could be monitor in future research.

Long-term food deprivation redounded to a significant increase in plasma cholesterol and triglyceride concentrations with a greater level in starved fish versus restricted-fed fish. In white sturgeon and beluga, similar to our findings, plasma triglyceride was mobilized during starvation (Falahatkar, 2012; Hung et al., 1997). Reduction in plasma cholesterol concentrations was found in rainbow trout and Adriatic sturgeon at 5 and 40 days of starvation respectively (Furné et al., 2012). A significant reduction in total cholesterol and triglyceride concentrations also found in carp *Cyprinus carpio* and Antarctic fish *Notototheni coriiceps* (Friedrich & Stepanowska, 2001; Stepanowska et al., 2006). Shimeno, Shikata, Hosokawa, Masumoto, and Kheyyali (1997) reported a reduction in plasma and body composition lipids as well as the enzyme activities related to the feeding rates in common carp, but similar result was not found in Siberian sturgeon (Jafari et al., 2018). In contrast, no changes have been shown in plasma triglyceride and cholesterol concentrations in Atlantic salmon, 2 and 14 days after starvation (Hevrøy et al., 2011). Plasmatic triglyceride concentration is the result of de novo hepatic lipogenesis and very low-density lipid secretion and peripheral tissue removal. It is demonstrated that under feed deprivation, the lipid synthesis depressed by rising β -oxidation, as a result of reducing essential enzymes such as glucose 6-phosphate dehydrogenase and malic enzyme (Pérez-Jiménez et al., 2012). In some species, rising of liver 3-hydroxy-CoA-dehydrogenase activity and triglyceride levels and simultaneous decline of plasma triglyceride level have indicated an increment of lipid metabolism for the periods without food (Polakof, Ceinos, Fernández-Durán, Míguez, & Soengas, 2007). In fact, in the present study, increase in plasma lipid and triglyceride levels in concomitant with protein concentration suggested that the lipid substrates were used parallel with protein deposits even in the late stage of starvation. These reactions may be due to consecutive use of energy sources during food deprivation period.

According to the results, although starvation in most cases had significant effects on juvenile Siberian sturgeon, restricted-fed fish compared with the control did not impose considerable impacts on body composition, biochemical and haematological parameters. Nevertheless, since most of the possible effects of restricted feeding were observed at the end of trial, it seems that Siberian sturgeon has the ability to adjust the metabolic rate in short-term (4 weeks) limited-fed period with using depleted stores first from readily available reserves. Finally, given the above-mentioned and also FCR improvement during the trial, the results clearly indicate the benefits of restricted feeding in relation to growth and physiological parameters and revealed that Siberian sturgeon is able to tolerate and compatibility with this condition. Therefore, it can be expressed that this experimental condition is advising as a management strategy in a short period to be used in Siberian sturgeon production when the rearing condition is unfavourable. Nonetheless, more research is needed to express how restricted feeding may affect muscle and liver tissue changes (in terms of composition and histological assay), immune system, stress responses, enzymatic activity, gene expression of responsible hormones and pathway mechanism.

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